PHANTOMSPECTRE

WiFi + cell survey console for Android · install, first walk, layers, troubleshooting

What you need

- An Android phone on the WiFi you want to map. Root is not required.
- **Termux** and the **Termux:API** add-on app, both from F-Droid (the Play Store builds are outdated and do not work together).
- The two files: `phantom.py` and `phantom.html`, saved in the same folder.

1 · Install

Open Termux and run:

```
pkg update && pkg upgrade
pkg install python termux-api
termux-setup-storage
```

Then open the Termux:API app once and allow **Location** and **Phone** when asked (Phone is for cell signal). Android gates WiFi scan results behind location permission; without it the listener sees nothing. On Android 10+ also turn Location on in the quick-settings panel.

2 · Put the files in place

Download the two files to the phone, then move them into a folder inside Termux:

```
mkdir -p ~/phantom
cp ~/storage/downloads/phantom.py ~/storage/downloads/phantom.html ~/phantom/
cd ~/phantom
```

If the download went to a different folder, change the path. `ls ~/storage/downloads` lists what is there.

3 · Start the listener

```
cd ~/phantom
termux-wake-lock
python phantom.py
```

You should see `PHANTOM SPECTRE listener on http://127.0.0.1:8765` and a line saying which tools it found. Leave this Termux session running. `termux-wake-lock` stops Android from killing it when the screen is off.

4 · Open the app

In Chrome on the same phone open:

```
http://127.0.0.1:8765
```

The status pill turns **WALK** and the gauge shows live dBm. If it stays OFFLINE, go to **Setup** → **Test connection**; the message there says what is wrong. Chrome menu → **Add to Home screen** installs it as a full-screen app with the Spectre icon.

5 · Walk your first survey

01 Survey tab is in **Walk** mode by default. Tap **Centre me**, or tap the map where you actually are.

02 Hold the phone flat, top edge forward, and press the arrow (↑ → ↓ ←) matching the map direction you face.

03 Walk. A reading logs itself every 0.8 m with WiFi dBm, cell dBm and ping. The map follows you; the ✦ marks the best spot and the guide arrow points to it.

04 Drifted? Tap your real position. Steps too long or short? Adjust **Stride**.

05 Switch layers: **WiFi**, **Cell**, **Ping**, **AP** (one access point) or **Best AP** (which AP wins where). **3D** shows the layer as terrain.

06 **Setup** → **JSON** or **CSV** → Copy, and paste into a note. Data lives in the browser only.

Reading the numbers

dBm	Meaning
`-30 to -50`	Excellent. Next to the router.
`-50 to -60`	Good. Default **Good** threshold is -60.
`-60 to -70`	Usable for video calls and streaming. Default **Usable** threshold is -70.
`-70 to -80`	Weak. Pages load, calls drop. Needs a repeater or AP move.
`below -80`	Dead zone for practical purposes.

Live devices and fences

The **Devices** tab shows every device on the WiFi in real time. Every 15 s the listener pings all 254 addresses and reads the ARP table; ARP is answered by the WiFi chip even when a phone is asleep, so this works without root. A device is **gone** after 3 missed sweeps (about a minute). Name devices and tap the bell to watch them.

Fence	What it does
`Network fence`	The WiFi itself is the fence. When a device drops off (walks out of coverage, switches off), you get an Android notification through Termux even with the app closed. Choose all devices or watched only, and whether joins notify too.
`Map fence`	A length × breadth rectangle on the active map, checked against your own walk-tracked position. Alerts when you step out and when you come back. Use **Centre on me** then Save.

A phone cannot tell where *other* devices are, only whether they are on the network. That is why the network fence is presence-based.

Tap **Enable alerts on this phone** once so Chrome can show notifications while the app is open. Listener notifications need the Termux:API app; Termux itself must be excluded from battery optimisation or Android will kill the sweep.

Troubleshooting

Symptom	Fix
`Status stays OFFLINE`	Page must be opened from http://127.0.0.1:8765 on the phone running phantom.py. The hosted claude.ai copy cannot reach the listener; use it for Manual mode and viewing imports only.
`Walk tracker says no sensors`	termux-sensor needs the Termux:API app. Run `termux-sensor -l` to list sensors; without a rotation vector sensor heading stays fixed and you press an arrow at each turn.
`CELL shows —`	Grant Termux:API the Phone permission. Some phones report nothing without a SIM.

Symptom	Fix
`Devices tab is empty`	Make sure the phone is on WiFi; the listener pings the whole /24 and reads the ARP table. First results take one sweep (15 s).
`termux-api not installed`	`pkg install termux-api` and install the Termux:API app from F-Droid.
`rssi is -127 or no wifi link`	Phone is not on WiFi, or Termux:API lacks Location permission. Grant it in Android Settings → Apps → Termux:API → Permissions.
`Scan now returns nothing`	Android allows 4 scans per 2 minutes. Wait, or rely on the live link only. Location toggle must be on.
`Listener dies when screen locks`	Run `termux-wake-lock` before starting, and exclude Termux from battery optimisation.
`Readings saved but hidden`	Map is bound to a different SSID than the one you are connected to. Unbind in Networks, or connect to the bound network.
`Data gone after reinstall`	Browser storage was cleared. Import your last export on the Setup tab.

Listener API

Useful if you want to log from another script or feed the data elsewhere.

Route	Returns
`GET /`	phantom.html
`GET /api/track`	position, heading, steps, wifi {ssid,bssid,rssi,…}, cell {dbm,type,operator}, net {latency_ms,gateway,ip}, motion
`GET /api/scan`	{ items: [{ ssid, bssid, rssi, frequency_mhz }] } cached 25 s
`GET /api/devices`	{ items: [{ key, ip, mac, host, name, watch, online, age_s, joins }], events, cfg } ?since=seq
`GET /api/notify`	?title=&body;= posts an Android notification via termux-notification
`GET /api/status`	tools found, sensors, operator, gateway
`GET /api/track/set?x=&y;=`	relocate; /face?deg=, /steplen?m=, /reset

```
curl -s http://127.0.0.1:8765/api/track | python -m json.tool
curl -s http://127.0.0.1:8765/api/devices
```

Options

Setting	Effect
`PHANTOM_PORT=9000 python phantom.py`	Change the port. Update Listener URL on the Setup tab to match.
`PHANTOM_HOST=0.0.0.0 python phantom.py`	Expose on the LAN so another device can open the app at http://<phone-ip>:8765. Readings still come from the phone's radio.
`Setup → Poll interval`	How often the gauge refreshes. 1000 ms is fine; 500 ms for a faster sparkline.
`Setup → Heat resolution`	Fine looks smoother but is slower on large maps with many readings.

Setting	Effect
`Setup → Keep polling when hidden`	Keep the sparkline running while the tab is in the background.

Autostart on Termux launch

Add this to the end of `~/.bashrc` so the listener starts whenever Termux opens:

```
if ! pgrep -f phantom.py >/dev/null; then
(cd ~/phantom && termux-wake-lock && nohup python phantom.py >/dev/null 2>&1 &)
echo 'phantom listener started'
fi
```

Positioning is dead reckoning: step counter plus rotation sensor. It drifts a little per minute of walking, so tap your true position now and then. The phone cannot locate other people from WiFi; the Devices tab lists who is on the network, the Motion card detects something crossing the phone–router path, and that is the honest limit of a phone radio.